

The Exponent Rules (Homework)

Simplify each expression below and write the Roman numeral of the exponent rule that you used in the margin.

① $(x^4)^9$

② 5^{-3}

③ $\frac{a^{15}}{a^{10}}$

④ $\left(\frac{2}{b}\right)^4$

⑤ 7^0

⑥ $36^{\frac{1}{2}}$

⑦ 2^{-2}

⑧ 34^1

⑨ $a^{10} \cdot a^{11}$

⑩ $(-64)^{\frac{2}{3}}$

⑪ $(8y)^0$

⑫ $(-9x)^2$

⑬ $\frac{b^8}{b^4}$

⑭ $\left(\frac{-8}{x}\right)^2$

⑮ b^1

★ ⑯ $\sqrt{100^3}$

⑰ $(10ab)^2$

⑱ $(b^6)^4$

⑲ $(-8)^{\frac{1}{3}}$

★ ⑳ $\sqrt[3]{125^2}$

㉑ $y^8 \cdot y^2 \cdot y^4$

$$(22) (1000)^{\frac{2}{3}}$$

$$(23) 8^{-2}$$

$$(24) (-6)^0$$

$$(25) 16^{\frac{1}{2}}$$

$$(26) b^2 \cdot b^{12}$$

$$(27) (8x)^2$$

$$(28) \left(\frac{1}{2}\right)^0$$

$$(29) 10^1$$

$$(30) \frac{a^8}{a^3}$$

$$(31) 3^{-4}$$

$$(32) (c^4)^4$$

$$(33) \left(\frac{c}{10}\right)^2$$

$$\star (34) \sqrt[4]{812}$$

$$(35) y^{23} \cdot y^{12}$$

$$(36) \frac{b^{10}}{b^7}$$

$$(37) (x^8)^3$$

$$(38) 256^{\frac{1}{4}}$$

$$\star (39) \sqrt{16^4}$$

$$(40) 8^{\frac{2}{3}}$$

$$(41) (-3xy)^2$$

$$(42) \left(\frac{5}{y}\right)^1$$

$$(43) \left(\frac{3}{c}\right)^2$$

$$(44) 16^{\frac{5}{4}}$$

Answers

V ① x^{36}

VIII ② $\frac{1}{125}$

II ③ a^5

IV ④ $\frac{16}{b^4}$

VII ⑤ 1

IX ⑥ 6

VIII ⑦ $\frac{1}{4}$

VI ⑧ 34

I ⑨ a^{21}

X ⑩ 16

VII ⑪ 1

III ⑫ $81x^2$

II ⑬ b^4

IV ⑭ $\frac{64}{x^2}$

VI ⑮ b

X * ⑯ 1000

III ⑰ $100a^2b^2$

V ⑱ b^{24}

IX ⑲ -2

X * ⑳ 25

I ㉑ y^{14}

$$X \text{ (22) } 100$$

$$VIII \text{ (23) } \frac{1}{64}$$

$$VII \text{ (24) } 1$$

$$IX \text{ (25) } 4$$

$$I \text{ (26) } b^{14}$$

$$III \text{ (27) } 64x^2$$

$$VII \text{ (28) } 1$$

$$VI \text{ (29) } 10$$

$$II \text{ (30) } a^5$$

$$VIII \text{ (31) } \frac{1}{81}$$

$$V \text{ (32) } c^{16}$$

$$IV \text{ (33) } \frac{c^2}{100}$$

$$X \star \text{ (34) } 9$$

$$I \text{ (35) } y^{35}$$

$$II \text{ (36) } b^3$$

$$V \text{ (37) } x^{24}$$

$$IX \text{ (38) } 4$$

$$X \star \text{ (39) } 256$$

$$X \text{ (40) } 4$$

$$III \text{ (41) } 9x^2y^2$$

$$VI \text{ (42) } \frac{5}{y}$$

$$IV \text{ (43) } \frac{9}{c^2}$$

$$X \text{ (44) } 32$$